ShopForHome

Comprehensive E-commerce Platform

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Batch:
WIPRO NGA - .Net Full Stack Angular - FY26 - C4
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Chapter 1

Problem Definition and Objectives

1.1 Problem Statement

The rapid growth of digital technologies has transformed the way consumers interact with retail systems. Traditional shopping methods often involve physical visits, limited product availability, and lack of personalization.

The ShopForHome platform provides a scalable and secure e-commerce solution for home products.

1.2 Objectives

- Enhance user experience
- Ensure secure authentication
- Develop admin control system
- Improve performance
- Maintain data integrity

Chapter 2

Frontend & Backend Architecture

2.1 Technology Stack

Frontend:

- Angular 21
- Bootstrap 5, Tailwind CSS

Backend:

- ASP.NET Core Web API
- Entity Framework Core
- JWT Authentication

Database:

- SQL Server

2.2 Architecture

Three-tier architecture:

Angular → ASP.NET Core → SQL Server

Chapter 3

Component Breakdown & API Design

3.1 Frontend Components

- Home, Products, Product Details
- Login, Register
- Cart, Wishlist, Orders
- Admin Dashboard, Upload, Reports

3.2 Backend APIs

- UsersController
- ProductsController
- CategoriesController
- OrdersController
- CartsController
- BulkUploadController
- SalesReportController

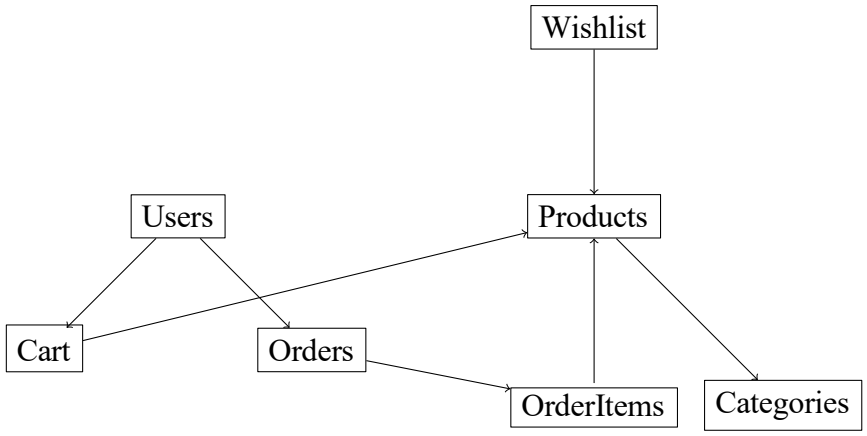
3.3 Authentication

JWT based authentication with role-based access.

Chapter 4

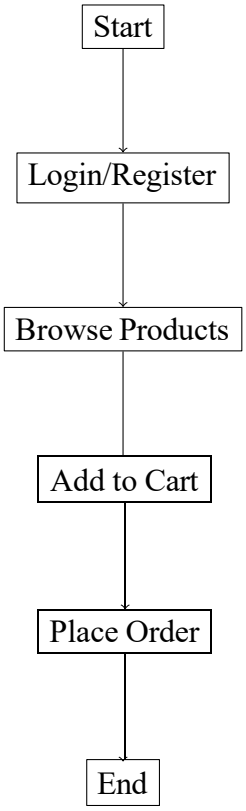
Database Design & ER Diagram

4.1 ER Diagram



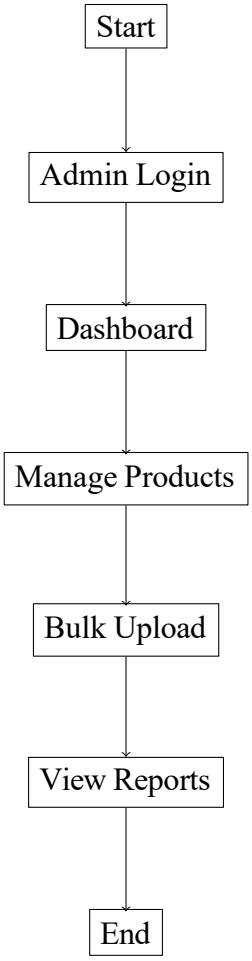
Chapter 5

User Flowchart



Chapter 6

Admin Flowchart

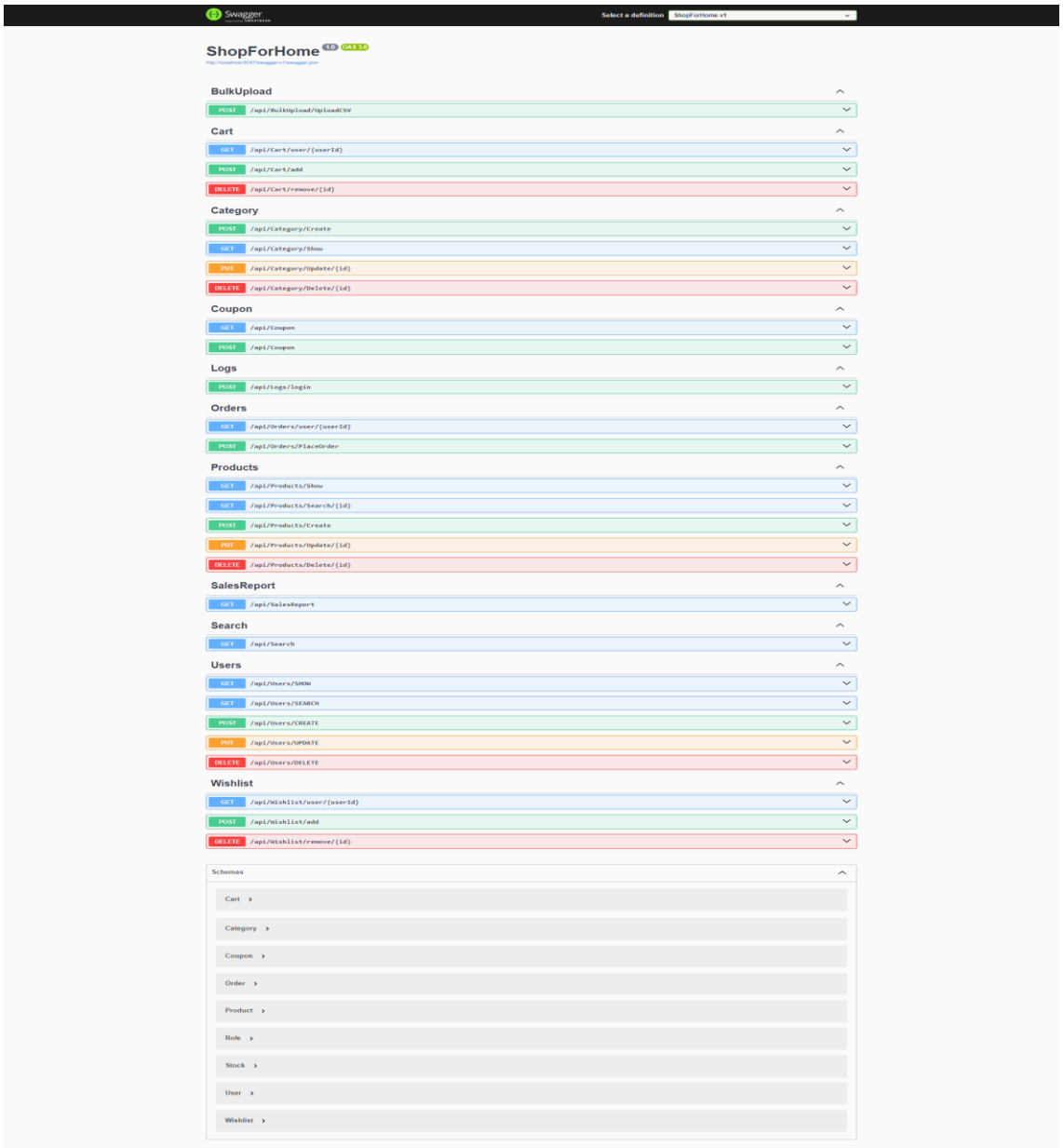


Chapter 7

Demonstration & Supporting Files

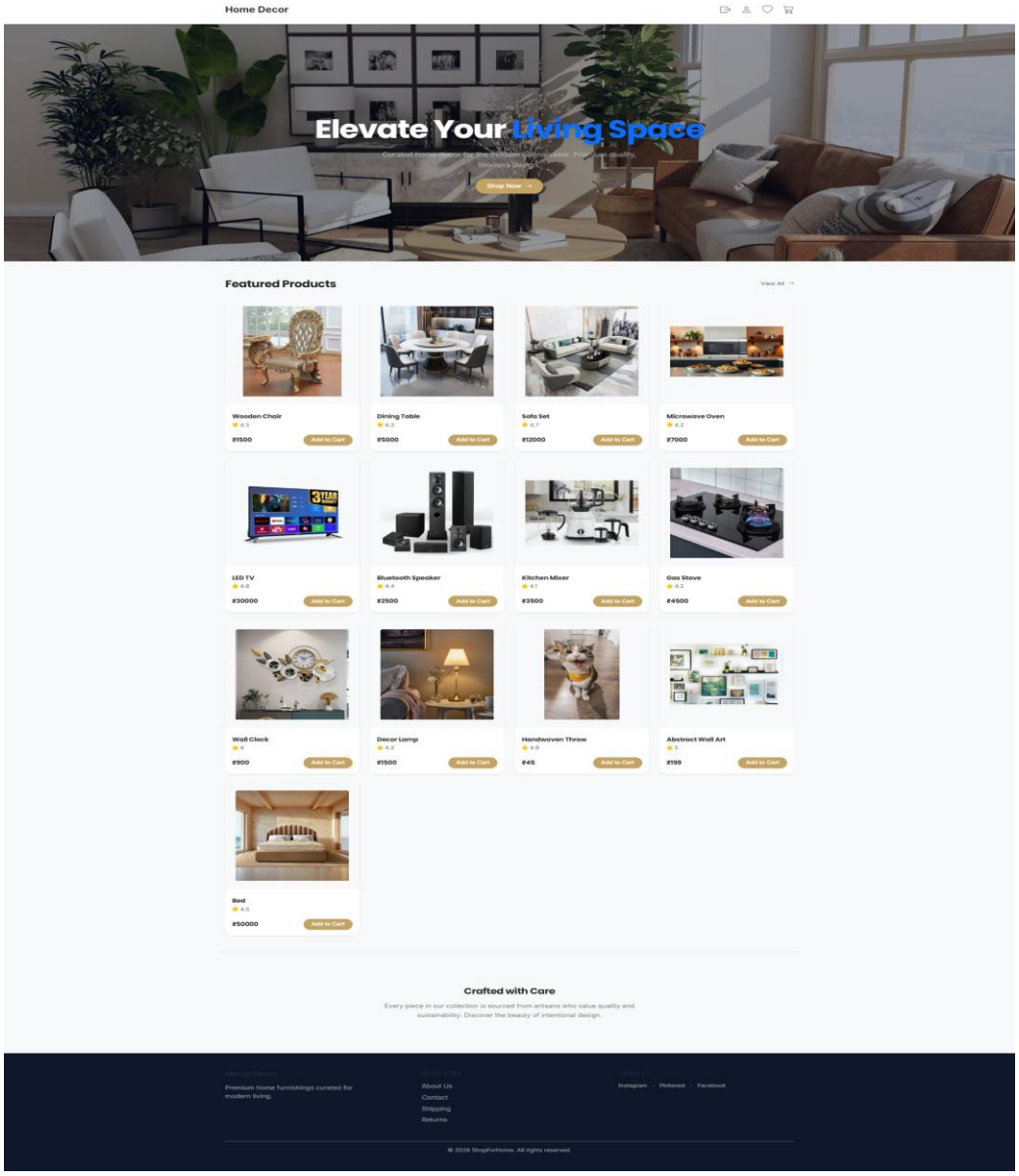
7.1 Files Demo

Backend:



Chapter 7

Front End:



Chapter 7

7.2 Files Included

7.2.1 Frontend and Backend Repository:

https://github.com/yeswanth088/Final_Capstone_wipro

- **Appsettings.json:**

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "ShopCon": "Server=DESKTOP-U2PO02N\\SQLEXPRESS;Database=ShopForHomeDB;Trusted_Connection=True;TrustServerCertificate=True"
  },
  "Jwt": {
    "Key": "Yeswanth@123",
    "Issuer": "ShopForHome",
    "Audience": "ShopForHomeUsers"
  }
}
```

GitHub Repository:

https://github.com/yeswanth088/Final_Capstone_wipro

Chapter 8

Deployment

8.1 Execution

Backend: dotnet run
Frontend: npm start

8.2 Deployment

Deployed via GitHub and VS Code.

Chapter 9

Conclusion

ShopForHome project successfully delivers a scalable and secure e-commerce platform that meets modern user expectations. By leveraging advanced technologies such as Angular, ASP.NET Core, and SQL Server, the system ensures high performance, maintainability, and user satisfaction.

The project demonstrates strong understanding of full-stack development, API design, database management, and secure authentication mechanisms, making it a robust and industry-relevant solution.